4.6.1 Pedagogy for Science

Learning Science involves not just learning theories and facts of Science, but also making connections between conceptual learning and real life, acquiring the capacities of scientific inquiry, and most importantly, applying these to understanding the world.

Students like to explore the world around them and understand why and how things happen. In this process of exploration, they use trial-and-error methods to test their hypotheses and reach possible conclusions. This exploration need not take place individually - students learn Science best through engaging with peers and adults.

Students have theories about why things happen, patterns they see around them, and about cause-and-effect relationships. As they learn about Science in a more formal set-up, these ideas get tested. Some concepts fit into the students' current understanding, while others require a shift in thinking. If there is an alignment between current ideas and what is discussed in classrooms, ideas get strengthened.

At the same time, some concepts do not fit into the students' current thinking. If not addressed, they can turn into alternative conceptions. For example, heavy objects fall faster, plants are non-living because they don't move, or heavy/ big objects always sink in water. If these ideas are not challenged and suitably modified through investigation, they can turn into alternative conceptions which may persist as students move through school.

Apart from these theories, students also bring with them the ability to reason, understand, and explain relationships between cause and effect. These capacities serve as the basis for developing scientific reasoning. Opportunities, therefore, to inquire are important, as opposed to being 'told'.

Scientific values, such as honesty and integrity, also develop through 'doing Science'. For example, while demonstrating an experiment on the boiling point of water, we should write the reading on the thermometer accurately, even if the water is not boiling at 100 degrees.

The role of the Teacher in aligning pedagogy and assessment with how students learn Science is critical. Teachers must build an environment that promotes natural curiosity, encourages questions, gives maximum possible opportunities for hands-on activities, and gives ample space to discuss ideas. Opportunities for students to express their understanding through different modes and formative assessments to track growing understanding are also key to learning Science.

4.6.1.1 Pedagogic Principles

Science pedagogy across stages must be informed by the following principles:

a. Learning Science requires an active engagement of students with the world around them to understand it. Science pedagogy achieves this through:

   i. Simulating the processes of Science such as asking questions, hypothesising, observing, testing, finding evidence, collecting data, analysing, modifying conclusions, communicating, and re-questioning.

   ii. Exposing students to a variety of aspects of learning Science in varied settings — the laboratory, classroom, and field — through approaches such as inquiry, discovery, didactic, hands-on Science.

   iii. Encouraging and sustaining curiosity by providing varied experiences that may challenge students' existing notions and ideas.

b. Learning Science requires communication and sharing of ideas and observations. Science pedagogy achieves this through:

   i. Using scientific vocabulary during instruction and creating a variety of contexts and situations for students to communicate their understanding, ideas, and observations.

   ii. Peer and collaborative learning.

c. Learning Science requires a gradual increase in the capacity to engage with complex and abstract ideas, aligned with the cognitive and procedural capacities of students. Science pedagogy achieves this by building on students' existing knowledge and using multiple representations (mathematical, graphical, diagrammatic, and models).

d. Learning Science requires making linkages between knowledge for the holistic and multidisciplinary learning emphasised in NEP 2020. Science pedagogy achieves this through:

   i. Connecting scientific knowledge inside and outside the classroom.

   ii. Horizontal connections with other curricular areas.

e. Learning Science enables the development of certain values, such as collaboration, sensitivity, empathy, equality of opportunities, respect for diversity, and other values mentioned in NEP 2020. Science pedagogy must facilitate this process.

4.6.1.2 Recommended Pedagogical Approaches and Settings

The same pedagogical approach can be used across the three settings most suitable for learning Science - the classroom, the field, and the laboratory. This section recommends pedagogical approaches across a variety of settings in detail. The following is a list of considerations based on which Teachers can choose pedagogical approaches and settings:

a. The nature of a concept should guide the decision regarding the approach and setting. For example, speed can be discussed in the playing field, but the structure of a cell requires a microscope.

b. The approach and setting chosen should enable the attainment of learning outcomes and competencies.

c. Each of the recommended approaches and settings must be selected at least once in an academic year, if not more. This will ensure exposure to varied approaches and settings.

d. Even when Teachers choose a didactic approach, areas that students could have potentially inquired about or discovered should be highlighted.

a. Hands-on Science

The most important part of learning Science is actually 'doing Science' through hands-on experiential learning. 'Doing Science' can range from trial and error, using materials around them, or using basic scientific instruments (measuring instruments), and laboratory apparatus. In this process, students gain conceptual understanding and develop capacities through manipulating, designing, and building experiments and demonstrations.

b. Discovery approach

Students explore the natural world following their own interests and discover patterns of how the world works during their explorations. Teachers may also create opportunities or draw attention to natural phenomena that students can explore further. Often, this discovery is followed by other, more structured approaches to ensure learning. For example, the Teacher draws the attention of the students to changes in the length of the shadows as the day progresses or to the venation patterns of the leaves of different plants. Students' observations on shadows are then connected to scientific concepts such as the path of light, and the venation pattern is connected to the shapes of the leaves.

c. Inquiry approach

The inquiry approach allows students to navigate through unknown questions and explore solutions on their own. It allows students to work in the same way as scientists. The inquiry approach engages students with systematic observation, visualising, experimenting, inferring, communicating, and discovering relations. This approach allows Teachers to choose the appropriate type of inquiry with respect to the concept and to scaffold (support as per need) students' learning. For example, students could explore questions such as: How do the image characteristics vary with the relative position between lens and object? How does the surface area of the reactants affect the rate of reaction? How does the intensity of light affect the rate of photosynthesis?

d. Project-centred approach

This approach allows learning within the classroom to continue outside the classroom and extend over a period of time. For example, observing the changes in the moon over a month to understand the phases of the moon. In this process, connections to daily life are also made. The project-centred approach allows students to develop artefacts/products (charts, presentations, speeches) that reflect and communicate their emerging understanding. It also allows the integration of concepts across different curricular areas. For example, visits to the sites of local professional communities and interactions with the people engaged there, such as potters, weavers, craftspersons, farmers, blacksmiths, carpenters, and electricians would enable integrating concepts from vocational education and art with Science.

e. Didactic approach

Often, teaching Science involves communicating certain important information in the form of scientific terms, phenomena, and the historical development of concepts and ideas. In this approach, the Teacher largely regulates the direction and flow of the lesson. For example, after students have discovered changes in the length of the shadows throughout the day, the Teacher can explain the effect of the position of the sun on the length of the shadow and how students can use it to keep track of the time as well.

f. Demonstration

The Teacher demonstrates the working of certain instruments or outcomes of experimental set-ups to draw the attention of the students to relevant concepts. These demonstrations enrich students' learning experiences of the concepts.

These approaches can be implemented in a variety of settings. A combination of the recommended pedagogical approaches and settings is also often needed to be used for teaching a concept.

The box below illustrates a Science class that uses some of these pedagogical approaches within a laboratory setting in the voice of a Teacher.

Teacher's Voice 4.6i

Demonstrating Osmosis in Living Cells

For my Grade 9 students, I planned a demonstration of osmosis in living cells. They already are familiar with the concept in form of daily life examples such as what causes pruning of our fingers if kept in water for long, but in this session, we go a level deeper to observe the process in living cells under the microscope.

We will have a total of 90 minutes to perform this. Hence, I prepare the material such as common salt (NaCl) solution and cut onion pieces and equipment such as glass slides, forceps, and microscopes. Students work in multiple groups. Hence, I prepare as many instruction/observation recording sheets.

On that day, I begin by demonstrating the experiment — I prepare three slides A, B and C, add drops of solution in varying quantity, place coverslip, and then observe under the microscope what is happening to the cells in the onion peel on the slide.

Then all groups repeat this same set of actions and record their observation by drawing on the sheet provided. Post this, we discuss whatever we observed. After the students discuss possible inferences, I introduce the term 'osmosis' to the class, using the concept to explain their observations and share more daily life examples where we see this phenomena — for example, in the process of pickling.

4.6.1.3 Horizontal Connections

a. Horizontal connections with other curricular areas are necessary for the holistic and multidisciplinary learning emphasised in NEP 2020. Some curricular goals and competencies in both the Middle and Secondary Stage are designed to ensure horizontal connections between Science and other curricular areas. At the same time, pedagogy must be designed so that these connections are made in the classroom.

b. Pedagogic approaches and methods such as inquiry and projects, by their nature, provide scope to utilise concepts and capacities that cut across the disciplines of Science — for example, a project on investigating the sound produced by different musical instruments, and how this sound can be varied. Qualities and properties of sound produced both in terms of aesthetics, physics concepts involved, mathematical patterns, and human perception lead to a holistic appreciation and integration of competencies across curricular areas.

c. Pedagogic methods such as surveys and field-based methods enable students to see concepts through socio-cultural, economic, emotional, and scientific lenses — for example, a survey of traditional medicinal and cooking practices and their connection with the seasons.

4.6.1.4 Resources in Science Teaching

Science laboratories are essential for a good Science Education class. However, there are no separate rooms for Science laboratories in the majority of Middle schools currently, although

Science kits are provided. In this situation, Teachers can use their classrooms or any open space for conducting experiments. The following must inform the use of resources:

a. The materials and equipment should be simple and easy to use. This makes it more likely that they will be used in classrooms by Teachers. At the Middle Stage, Science kits available at most schools provide a good start.

b. However, teaching should not be restricted to the Science kits. The more materials students use, the more opportunities they get to do Science and hence learn Science. Improvised apparatus can be made using inexpensive materials to extend the use of materials beyond the Science kit. For example, a measuring jar can be made out of discarded transparent glass bottles, - using measuring cups (that usually come with syrup bottles) and syringes for calibration.

c. At this Stage, if the school can provide dedicated lab space, with adequate space for simple materials and resources, it must be done.

d. At the same time, doing Science must not be restricted to Science laboratories or Science kits. Classrooms, especially in the Middle Stage, must allow the doing of Science. At the same time, all safety considerations must be kept in mind.

e. Tinkering laboratories – informal spaces where students can 'play' with simple scientific materials and equipment independently – can be set up in any room within the school. This will help students strengthen design thinking, creating, and experimental capacities. Initially, students would have to be supported by the Teacher.

f. Students at the Secondary Stage would require standard scientific equipment and apparatus and basic infrastructure in which they can perform experiments with convenience and safety. Therefore, Secondary schools should have well-equipped, resourceful, and spacious Science laboratories to conduct Science experiments and investigations.

g. If a school has a laboratory but the number of students is large, the Teacher can either allow students to do the experiments in groups or ask students to perform the experiments on alternate days.

h. Budgets for Science in the Middle and Secondary Stages are limited, so Science equipment and materials should be inexpensive. However, if the equipment is of inferior quality (e.g., weak magnet, cheap microscope with plastic lens), it may not be worth using.

i. Alternatives can be used. For example, in case of unavailability of litmus paper, a Teacher can use turmeric solution or turmeric paper strips for identifying the acidic and basic characteristics of the substances. For this, the Teacher can take turmeric (powder or solid) and add it on a paper or in a glass cup containing water. This solution can be used for the identification of acids and bases. The Teacher can also make wet paper strips dipped in turmeric solution. Students can be asked to do the following — dry these paper strips, prepare solutions of each substance in water, dip the strip in the solution, and check the colour change of the turmeric paper strips. Could you make a list of changes in the colours of these substances?